02-719 Week 8

Genomics and Epigenetics of the Brain

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Recap: Convergent Evolution

- The same phenotype evolves **independently** in separate lineages
- Each origin is a natural replicate of the same experiment
  - If a gene matters for the trait, it may respond the same way each time
- The question: Does a gene's **rate of evolution** shift on the branches where are trait evolved?

Species are not independent data points

Naive approach:

- Correlate a gene's sequence with the trait across species
- Closely related species share both, so one origin can look like many
- $\rightarrow$ inflated significance

Phylogenetic approach:

- Work on branches of the tree, not species
- Count each independent origin once
- $\rightarrow$ signal must repeat across lineages

RER Coverage

RER = Relative evolutionary rates. **RERcoverage** is an R workflow that does the following analysis:

- Inputs: gene trees with a fixed topology
- Relative evolutionary rates (RERs)
- Encoding the phenotype (binary vs continuous traits)
- Testing for association
- From gene lists to biology

RERcoverage Workflow

1. Alignments: one per gene
2. Gene trees: fixed topology, gene-specific branch lengths
3. RERs: relative rate on every branch
4. Phenotype paths: trait mapped onto branches
5. Correlation: RER vs. trait, per gene
6. Enrichment: Pathways / gene ontology

Each gene gets one test. The output is a ranked list of genes whose rates shift with the trait across the tree.

Step 1: Inputs, one tree per gene

- Start from a multiple alignment of orthologs for each gene
- Fix the **species tree topology**; estimate only the branch lengths
  - Branch length = substitutions per site on that branch
  - Species missing the gene are pruned for that gene
- readTrees() combines all genes
  - Builds a **master tree**: average branch lengths across genes